**Project Design Phase-II**
**Technology Stack (Architecture & Stack)**

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|---|---|
| Date | 3 March 2026 |
| Team ID | EL2026TMID6719 |
| Project Name | MedSafe AI - AI-driven medical safety assistant |
| Maximum Marks | 4 Marks |

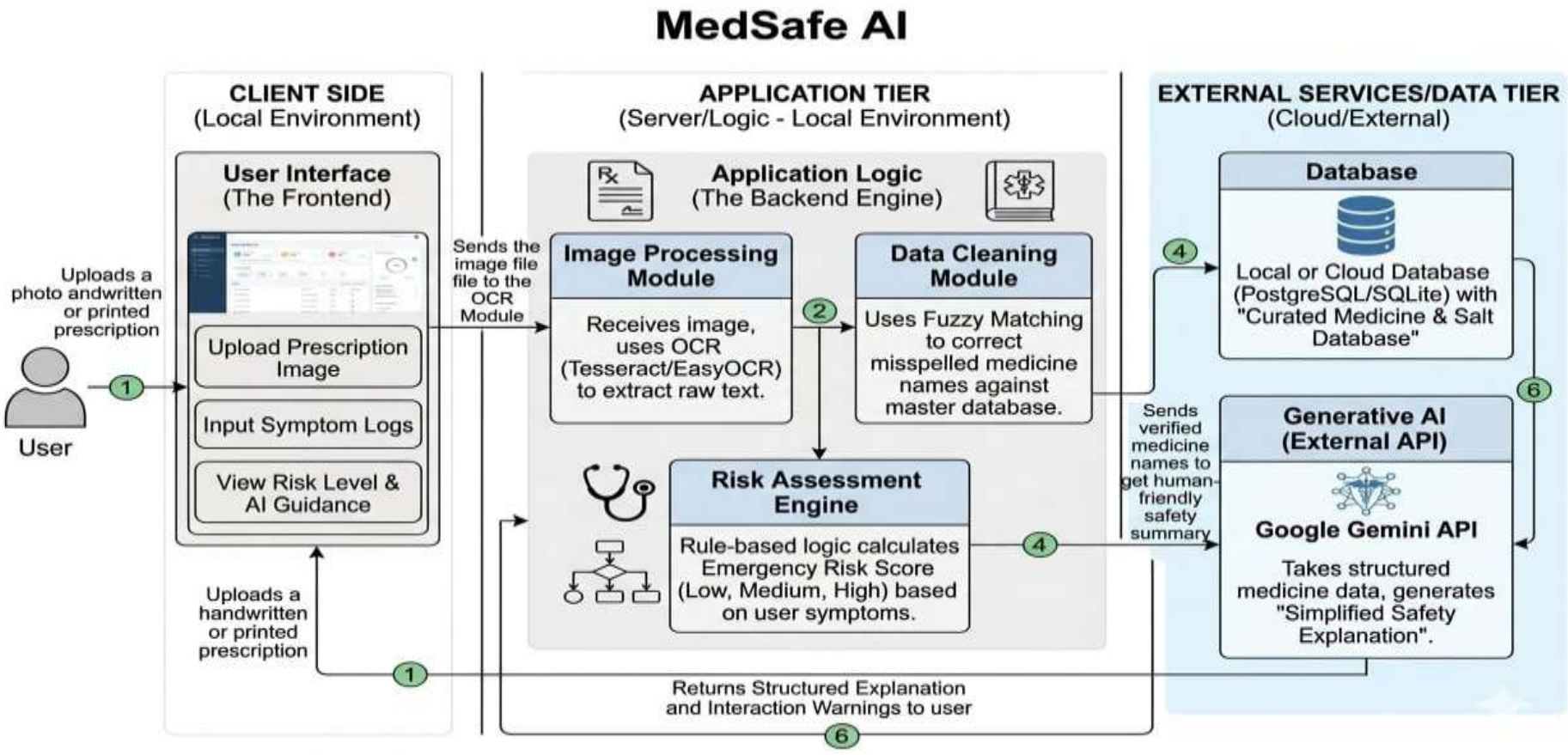

**Table-1: Technology Stack:**

| Sl. No. | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | Web-based dashboard for prescription uploads and symptom logging | Streamlit (Python-based UI) |
| 2. | Application Logic-1 | Core backend logic for OCR processing and medicine extraction | Python (Pytesseract / EasyOCR) |
| 3. | Application Logic-2 | Core backend logic for OCR processing and medicine extraction | Python (TheFuzz / Levenshtein) |
| 4. | Application Logic-3 | Risk scoring engine for emergency levels (Low/Med/High) | Python (Rule-based logic) |
| 5. | Database | "Storage for curated medicine names, active salts, and side effects." | SQLite or PostgreSQL |
| 6. | File Storage | Storage for uploaded prescription images for processing | Local Filesystem or Google Cloud Storage |
| 7. | External API-1 | AI engine for generating simplified safety explanations | Google Gemini API (Generative AI |
| 8. | Machine Learning Model | Model for identifying medicines and salts from OCR text | Large Language Model (LLM) |
| 9. | Infrastructure (Server / Cloud) | Environment where the application is hosted | Local (for dev) or Streamlit Cloud / Google Cloud Run |

**Table-2: Application Characteristics:**

| Sl. No | Component | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | Primary frameworks used for UI and backend | "Streamlit, Pandas, TheFuzz" |
| 2. | Security Implementation | Protecting user-uploaded data and API keys | "Environment Variables (.env), HTTPS encryption" |
| 3. | Scalable Architecture | The application is designed using modular functions for AI and DB | "3-tier Architecture (UI, Logic, Data)" |
| 4. | Performance | Rapid processing of OCR and AI responses for real-time feedback | Asynchronous API calls, Cached Database queries" |